The Phonon Symmetry

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Summary

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The Symmetry of Lattice Vibration

Let $r_\mu^{(ni)}$ be the μ-position vector ($\mu = x, y, z$) of i-th atom in n-th primitive cell, $V_{\mu\nu}^{(ni)(mj)}$ the second-order tensor. Then, we obtain a $3sN \times 3sN$ linear equation

$$m_i \ddot{r}_\mu^{(ni)} = -V_{\mu\nu}^{(ni)(mj)} r_\nu^{(mj)}. \tag{1}$$

Denote $r_\mu^{(ni)} = \frac{1}{\sqrt{m_i}} u_\mu^{(i)}(\mathbf{q}) \mathrm{e}^{\mathrm{i}\mathbf{q}\cdot\mathbf{R}_n - \mathrm{i}\omega t}$. Due to Bloch's theorem, (1) can be reduced to N $3s \times 3s$ equations

$$D(q)_{\mu\nu}^{(i)(j)}(\mathbf{q}) u_\nu^{(j)}(\mathbf{q}) = \omega^2 u_\mu^{(i)}(\mathbf{q}) \tag{2}$$

labeled by $\mathbf{q}$. Here $D(\mathbf{q})$ is so-called **dynamical matrix**. The solution set

$$\left\{ u_\mu^{(i)}(\mathbf{q}|j\lambda) \right\} \tag{3}$$

forms an orthonormal basis in (i, μ)-space.

Let $g = \{\alpha|t\} \in G$, and consider g acts on $u_\mu^{(i)}(\mathbf{q}|j\lambda)$. By completeness, we have

$$g \cdot u_{\mu_0}^{(i_0)}(\mathbf{q}|j\lambda) = u_\mu^{(\bar{i})}(\alpha \cdot \mathbf{q}|j\lambda) = \Gamma_{\mu\nu}^{\bar{i}i'}(\mathbf{q}|g)\, u_\nu^{(i')}(\mathbf{q}|j\lambda). \tag{4}$$

Proposition (**Tao**)

$$\Gamma(\mathbf{q}|g_1 g_2) = \Gamma(\alpha_2 \cdot \mathbf{q}|g_1)\Gamma(\mathbf{q}|g_2). \tag{5}$$

Here, Γ is so-called **mechanics representation**. And $\Gamma(\mathbf{q}|\bullet)$ forms a representation of little group $G_\mathbf{q}$. We irreducibly decompose $\Gamma(\mathbf{q}|\bullet)$ and each component is called **allowed phonon irreducible little representation**. The j-th linear subspace of $\left\{u_\mu^{(i)}(\mathbf{q}|j\lambda)\right\}$ affords one irreducible little representation.

Time-Reversal Symmetry

Non-magnetic materials naturally have time-reversal symmetry. When there is $g \in G$ so that $g \cdot \mathbf{q} = -\mathbf{q}$, we have

$$\Gamma(\mathbf{q}|g) D(\mathbf{q}) \Gamma(\mathbf{q}|g)^{-1} = D(-\mathbf{q}). \tag{6}$$

Due to $D(-\mathbf{q}) = \overline{D}(\mathbf{q})$, the time-reversal becomes a symmetry:

$$[\mathscr{C}\Gamma(\mathbf{q}|g), D(\mathbf{q})] = 0. \tag{7}$$

In this case, we need to study the **corepresentation** of "3-rd Shubnikov magnetic group"

$$G_{\mathbf{q},-\mathbf{q}} := G_{\mathbf{q}} \sqcup \mathscr{C} g G_{\mathbf{q}}. \tag{8}$$

Example: The Σ Line of Si

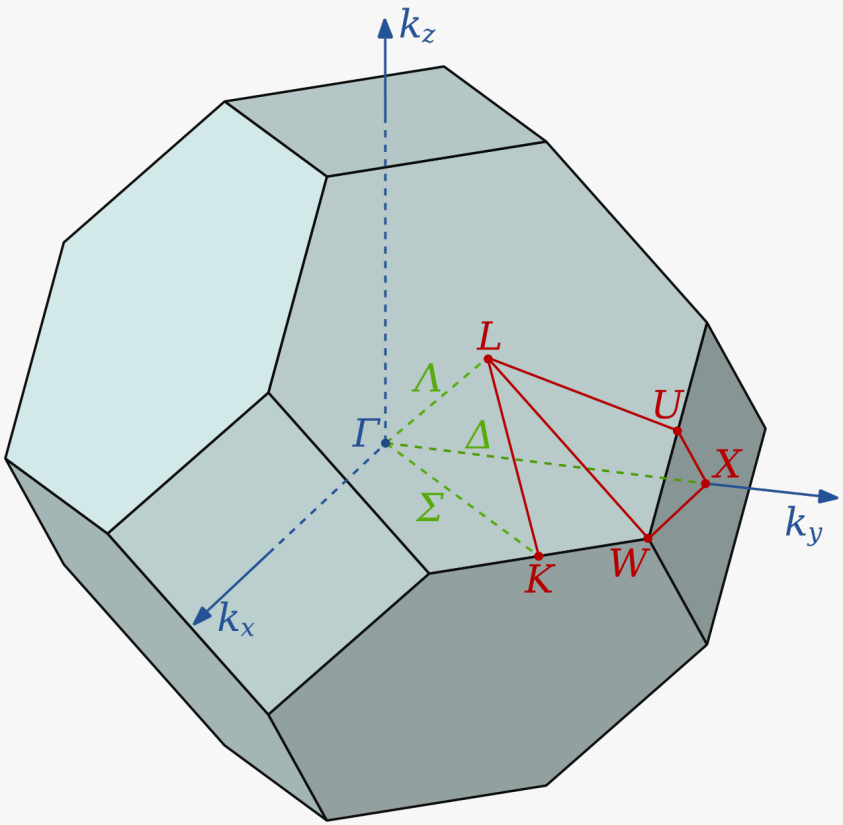


Figure 1: First Brilloutin zone of Si

The Σ line is along $[110]$ direction.

How about the phonon modes when $\mathbf{q} \in \Sigma$?

generators	real space: (x,y,z)	reciprocal space: (k_x,k_y,k_z)
$h_1=\{1\|0\}$	(x,y,z)	(k_x,k_y,k_z)
$h_{16}=\left\{C_{2,[110]}\middle\|\boldsymbol{\tau}\right\}$	$(y,x,\bar{z})$	$(k_y,k_x,\overline{k_z})$
$h_{28}=\left\{m_{(001)}\middle\|\boldsymbol{\tau}\right\}$	$(x,y,\bar{z})$	$(k_x,k_y,\overline{k_z})$
$h_{37}=\left\{m_{(1\bar{1}0)}\middle\|0\right\}$	(y,x,z)	(k_y,k_x,k_z)
$h_{25}=\{i\|\boldsymbol{\tau}\}$	$(\bar{x},\bar{y},\bar{z})$	$(\overline{k_x},\overline{k_y},\overline{k_z})$

Figure 2: The elements in $\overline{G}_\Sigma$ and their actions; $h_{25}\notin\overline{G}_\Sigma$

Due to $h_{25}:\mathbf{q}\mapsto-\mathbf{q}$, we need to consider time-reversal.

Let $\breve{T}(\mathbf{q})$ be the allowed little projective representation of mechanics little representation $\Gamma(\mathbf{q}|\bullet)$. According to **Mackey's theory**, there is a one-to-one corresponding between irreducible components of $\breve{T}(\mathbf{q})$ and $\Gamma(\mathbf{q}|\bullet)$, and by **character theory**, we have

$$\widetilde{T}(\Sigma) = 2\tau^{(1)} \oplus \tau^{(2)} \oplus \tau^{(3)} \oplus 2\tau^{(4)}, \tag{9}$$

which is

$$\begin{aligned}
\tau^{(1)}: &\quad \frac{1}{2}(1,1,0;\bar{\rho},\bar{\rho},0), \qquad \frac{1}{\sqrt{2}}(0,0,1;0,0,-\bar{\rho}),\\
\tau^{(2)}: &\quad \frac{1}{2}(1,-1,0;-\bar{\rho},\bar{\rho},0),\\
\tau^{(3)}: &\quad \frac{1}{2}(1,-1,0;\bar{\rho},-\bar{\rho},0),\\
\tau^{(4)}: &\quad \frac{1}{2}(1,1,0;-\bar{\rho},-\bar{\rho},0), \qquad \frac{1}{\sqrt{2}}(0,0,1;0,0,\bar{\rho}).
\end{aligned} \tag{10}$$

$$\widetilde{T}(\Sigma) = 2\tau^{(1)} \oplus \tau^{(2)} \oplus \tau^{(3)} \oplus 2\tau^{(4)}.$$

$$\tau^{(1)}: \quad \frac{1}{2}(1,1,0;\bar{\rho},\bar{\rho},0), \qquad \frac{1}{\sqrt{2}}(0,0,1;0,0,-\bar{\rho}),$$

$$\tau^{(2)}: \quad \frac{1}{2}(1,-1,0;-\bar{\rho},\bar{\rho},0),$$

$$\tau^{(3)}: \quad \frac{1}{2}(1,-1,0;\bar{\rho},-\bar{\rho},0),$$

$$\tau^{(4)}: \quad \frac{1}{2}(1,1,0;-\bar{\rho},-\bar{\rho},0), \qquad \frac{1}{\sqrt{2}}(0,0,1;0,0,\bar{\rho}).$$

So, $\tau^{(2)}$ and $\tau^{(3)}$ are pure TO and TA, respectively; $\tau^{(1)}$ and $\tau^{(4))}$ are LA+TO_z and LO+TA_z, respectively.

References

[1] R. Tao, *Group Theory in Physics*. Beijing: Higher Education Press, 2011, In Chinese, ISBN: 9787040312058.

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